Life Cycle Cost Analysis Report

3psLCCA Engine

April 12, 2026

Project Parameters

Parameter	Value
Service Life Years	75
Analysis Period Years	150
Discount Rate Percent	6.7
Inflation Rate Percent	5.15
Interest Rate Percent	7.75
Currency Conversion	88.73

Construction Cost Inputs

Construction Input	Value
Initial Construction Cost	12,843,979.44
Initial Carbon Emissions Cost	2,065,434.91
Superstructure Construction Cost	9,356,038.92
Total Scrap Value	2,164,095.02

1 Initial Stage

1.1 Summary

Component	Value
Economic: Initial Construction Cost	12,843,979.44
Economic: Time Cost Of Loan	215,671.82
Environmental: Initial Material Carbon Emission Cost	2,065,434.91
Environmental: Initial Vehicular Emission Cost	923,557.7
Social: Initial Road User Cost	20,064,546.22

1.2 Economic

Parameter	Value
Initial Construction Cost	12,843,979.44
Time Cost Of Loan	215,671.82

1.3 Environmental

Parameter	Value
Initial Material Carbon Emission Cost	2,065,434.91
Initial Vehicular Emission Cost	923,557.7

1.4 Social

Parameter	Value
Initial Road User Cost	20,064,546.22

2 Use Stage

2.1 Summary

Component	Value
Economic: Routine Inspection Costs	762,521.37
Economic: Periodic Maintenance	789,281.8
Economic: Major Inspection Costs	717,528.91
Economic: Major Repair Cost	2,920,720.92
Economic: Replacement Costs For Bearing And Expansion Joint	1,817,410.56
Environmental: Periodic Carbon Costs	126,924.07
Environmental: Major Repair Material Carbon Emission Costs	25,832.39
Environmental: Major Repair Vehicular Emission Costs	1,211,636.67
Environmental: Vehicular Emission Costs For Replacement Of Bearing And Expansion Joint	18,400.11
Social: Major Repair Road User Costs	26,323,141.21
Social: Road User Costs For Replacement Of Bearing And Expansion Joint	399,747.5

2.2 Economic

Parameter	Value
Routine Inspection Costs	762,521.37
Periodic Maintenance	789,281.8
Major Inspection Costs	717,528.91
Major Repair Cost	2,920,720.92
Replacement Costs For Bearing And Expansion Joint	1,817,410.56

2.3 Environmental

Parameter	Value
Periodic Carbon Costs	126,924.07
Major Repair Material Carbon Emission Costs	25,832.39
Major Repair Vehicular Emission Costs	1,211,636.67
Vehicular Emission Costs For Replacement Of Bearing And Expansion Joint	18,400.11

2.4 Social

Parameter	Value
Major Repair Road User Costs	26,323,141.21
Road User Costs For Replacement Of Bearing And Expansion Joint	399,747.5

3 Reconstruction

3.1 Summary

Component	Value
Economic: Total Demolition And Disposal Costs	425,135.72
Economic: Cost Of Reconstruction After Demolition	4,251,357.19
Economic: Total Scrap Value	716,315.45
Economic: Time Cost Of Loan	71,387.37
Environmental: Carbon Costs Demolition And Disposal	68,365.9
Environmental: Carbon Cost Of Reconstruction After Demolition	683,658.96
Environmental: Demolition Vehicular Emission Cost	58,788
Environmental: Reconstruction Vehicular Emission Cost	305,697.6
Social: Ruc Demolition	1,277,185.54
Social: Ruc Reconstruction	6,641,364.8

3.2 Economic

Parameter	Value
Total Demolition And Disposal Costs	425,135.72
Cost Of Reconstruction After Demolition	4,251,357.19
Total Scrap Value	716,315.45
Time Cost Of Loan	71,387.37

3.3 Environmental

Parameter	Value
Carbon Costs Demolition And Disposal	68,365.9
Carbon Cost Of Reconstruction After Demolition	683,658.96
Demolition Vehicular Emission Cost	58,788
Reconstruction Vehicular Emission Cost	305,697.6

3.4 Social

Parameter	Value
Ruc Demolition	1,277,185.54
Ruc Reconstruction	6,641,364.8

4 End-of-Life Stage

4.1 Summary

Component	Value
Economic: Total Demolition And Disposal Costs	142,568.17
Economic: Total Scrap Value	240,214.55
Environmental: Carbon Costs Demolition And Disposal	22,926.33
Environmental: Demolition Vehicular Emission Cost	19,714.4
Social: Ruc Demolition	428,300.89

4.2 Economic

Parameter	Value
Total Demolition And Disposal Costs	142,568.17
Total Scrap Value	240,214.55

4.3 Environmental

Parameter	Value
Carbon Costs Demolition And Disposal	22,926.33
Demolition Vehicular Emission Cost	19,714.4

4.4 Social

Parameter	Value
Ruc Demolition	428,300.89

5 Detailed Calculations

5.1 Initial Stage Detailed Calculations

5.1.1 Formulae

Parameter	Value
Initial Construction Cost	initial_construction_cost
Initial Material Carbon Emission Cost	initial_carbon_cost
Initial Road User Cost	calculated based on construction duration and daily road user cost
Initial Vehicular Emission Cost	calculated based on construction duration and daily vehicular emission cost
Time Cost Of Loan	calculated based on initial construction cost, loan interest rate, and construction period

5.1.2 Inputs

Parameter	Value
Initial Construction Cost	12,843,979.44
Initial Carbon Cost	2,065,434.91
Construction Period In Years	0.43
Days Per Month	30

5.1.3 Computed Values

Parameter	Value
Initial Road User Cost	20,064,546.22
Initial Vehicular Emission Cost	923,557.7
Time Cost Of Loan	215,671.82

5.1.4 Road User Cost Breakdown

Parameter	Value
Daily Ruc	128,618.89
Duration Days	156
Spwf Applied	1

5.1.5 Total Time Cost Of Loan

Time Cost Of Loan

This calculates time cost of loan.

Equation

$$C_{loan} = C_0 \times r \times t_c \times I \times SPWF$$

Substitution

$$C_{loan} = 12\,843\,979.44 \times 0.08 \times 0.43 \times 0.5 \times 1$$

Result

$$C_{loan} = 215\,671.82$$

Abbreviations

Symbol	Meaning
C_{loan}	Time Cost Of Loan
C_0	Initial Cost Of Construction
r	Interest Rate
t_c	Time For Construction Years
I	Investment Ratio
$SPWF$	Sum Of Present Worth Factor

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Interest Rate	0.08
Time For Construction Years	0.43
Investment Ratio	0.5
Sum Of Present Worth Factor	1

Computed Values

Computed Value	Result
Time Cost Of Loan	215,671.82

5.2 Use Stage Detailed Calculations

5.2.1 Routine Inspection Costs

Breakdown

Routine Inspection Cost Per Year

This calculates the annual routine inspection cost before discounting.

Equation

$$C_{ri,y} = \frac{C_0 \times p_{c,y}}{100}$$

Substitution

$$C_{ri,y} = \frac{12\,843\,979.44 \times 0.1}{100}$$

Result

$$C_{ri,y} = 12\,843.98$$

Abbreviations

Symbol	Meaning
$C_{ri,y}$	Routine Inspection Cost Per Year
C_0	Initial Cost Of Construction
$p_{c,y}$	Percentage Of Initial Construction Cost Per Year

Present Value Of Routine Inspection Costs

This converts routine inspection costs into present-value terms using the applicable discounting factor.

Equation

$$RIC = C_{ri,y} \times PWF$$

Substitution

$$RIC = C_{ri,y} \times PWF$$

Abbreviations

Symbol	Meaning
RIC	Present Value Of Routine Inspection Costs
$C_{ri,y}$	Routine Inspection Cost Per Year
PWF	Present Worth Factor

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Percentage Of Initial Construction Cost Per Year	0.1
Interval In Years	1

Computed Values

Computed Value	Result
Routine Inspection Cost Per Year	12,843.98
Sum Of Present Worth Factor	59.37

Sum Of Present Worth Factor Breakdown

Breakdown

Year To Pwf

Parameter	Value
1.43	0.98
2.43	0.96
3.43	0.95
4.43	0.94
5.43	0.92
6.43	0.91
7.43	0.9
8.43	0.88
9.43	0.87
10.43	0.86
11.43	0.85
12.43	0.83
13.43	0.82
14.43	0.81
15.43	0.8
16.43	0.79
17.43	0.78
18.43	0.76
19.43	0.75
20.43	0.74
21.43	0.73
22.43	0.72
23.43	0.71
24.43	0.7
25.43	0.69
26.43	0.68
27.43	0.67
28.43	0.66
29.43	0.65
30.43	0.64
31.43	0.63
32.43	0.62
33.43	0.61
34.43	0.6
35.43	0.59
36.43	0.59
37.43	0.58
38.43	0.57
39.43	0.56
40.43	0.55
41.43	0.55
42.43	0.54
43.43	0.53
44.43	0.52
45.43	0.51
46.43	0.51
47.43	0.5
48.43	0.49
49.43	0.48

Parameter	Value
50.43	0.48
51.43	0.47
52.43	0.46
53.43	0.46
54.43	0.45
55.43	0.44
56.43	0.44
57.43	0.43
58.43	0.42
59.43	0.42
60.43	0.41
61.43	0.41
62.43	0.4
63.43	0.4
64.43	0.39
65.43	0.38
66.43	0.38
67.43	0.37
68.43	0.37
69.43	0.36
70.43	0.36
71.43	0.35
72.43	0.35
73.43	0.34
74.43	0.34
76.87	0.33
77.87	0.32
78.87	0.32
79.87	0.31
80.87	0.31
81.87	0.3
82.87	0.3
83.87	0.29
84.87	0.29
85.87	0.28
86.87	0.28
87.87	0.28
88.87	0.27
89.87	0.27
90.87	0.27
91.87	0.26
92.87	0.26
93.87	0.25
94.87	0.25
95.87	0.25
96.87	0.24
97.87	0.24
98.87	0.23
99.87	0.23
100.87	0.23
101.87	0.23
102.87	0.22

Parameter	Value
103.87	0.22
104.87	0.22
105.87	0.21
106.87	0.21
107.87	0.21
108.87	0.2
109.87	0.2
110.87	0.2
111.87	0.2
112.87	0.19
113.87	0.19
114.87	0.19
115.87	0.18
116.87	0.18
117.87	0.18
118.87	0.18
119.87	0.17
120.87	0.17
121.87	0.17
122.87	0.17
123.87	0.16
124.87	0.16
125.87	0.16
126.87	0.16
127.87	0.15
128.87	0.15
129.87	0.15
130.87	0.15
131.87	0.14
132.87	0.14
133.87	0.14
134.87	0.14
135.87	0.14
136.87	0.14
137.87	0.13
138.87	0.13
139.87	0.13
140.87	0.13
141.87	0.12
142.87	0.12
143.87	0.12
144.87	0.12
145.87	0.12
146.87	0.12
147.87	0.12
148.87	0.11
149.87	0.11

Parameter	Value
Construction Period	0.43
Service Life	75
Interval	1

Parameter	Value
Total	59.37

Parameter	Value
Total Routine Inspection Costs	762,521.37

Parameter	Value
Total	762,521.37

5.2.2 Routine Maintenance And Carbon Costs

Breakdown

Routine Maintenance Cost Per Year

This calculates the annual routine maintenance cost before discounting.

Equation

$$C_{pm,y} = \frac{C_0 \times p_{c,y}}{100}$$

Substitution

$$C_{pm,y} = \frac{12\,843\,979.44 \times 0.55}{100}$$

Result

$$C_{pm,y} = 70\,641.89$$

Abbreviations

Symbol	Meaning
$C_{pm,y}$	Routine Maintenance Cost Per Year
C_0	Initial Cost Of Construction
$p_{c,y}$	Percentage Of Initial Construction Cost Per Year

Routine Carbon Cost Per Year

This calculates the annual routine carbon cost before discounting.

Equation

$$C_{pm,CO_2,y} = \frac{C_{CO_2,0} \times p_{CO_2}}{100}$$

Substitution

$$C_{pm,CO_2,y} = \frac{2\,065\,434.91 \times 0.55}{100}$$

Result

$$C_{pm,CO_2,y} = 11\,359.89$$

Abbreviations

Symbol	Meaning
$C_{pm,CO_2,y}$	Routine Carbon Cost Per Year
$C_{CO_2,0}$	Cost Of Initial Carbon Emissions In
p_{CO_2}	Percentage Of Initial Cabron Emission Cost

Present Value Of Routine Maintenance Costs

This converts routine maintenance costs into present-value terms using the applicable discounting factor.

Equation

$$RMC = C_{pm,y} \times PWF$$

Substitution

$$RMC = C_{pm,y} \times PWF$$

Abbreviations

Symbol	Meaning
RMC	Present Value Of Routine Maintenance Costs
$C_{pm,y}$	Routine Maintenance Cost Per Year
PWF	Present Worth Factor

Present Value Of Routine Carbon Costs

This converts routine carbon costs into present-value terms using the applicable discounting factor.

Equation

$$RCC = C_{pm,CO_2,y} \times PWF$$

Substitution

$$RCC = C_{pm,CO_2,y} \times PWF$$

Abbreviations

Symbol	Meaning
RCC	Present Value Of Routine Carbon Costs
$C_{pm,CO_2,y}$	Routine Carbon Cost Per Year
PWF	Present Worth Factor

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Cost Of Initial Carbon Emissions In	2,065,434.91
Percentage Of Initial Construction Cost Per Year	0.55
Percentage Of Initial Cabron Emission Cost	0.55
Interval In Years	5

Computed Values

Computed Value	Result
Routine Maintenance Cost Per Year	70,641.89
Routine Carbon Cost Per Year	11,359.89
Sum Of Present Worth Factor	11.17

Sum Of Present Worth Factor Breakdown

Breakdown

Year To Pwf

Parameter	Value
5.43	0.92
10.43	0.86
15.43	0.8
20.43	0.74
25.43	0.69
30.43	0.64
35.43	0.59
40.43	0.55
45.43	0.51
50.43	0.48
55.43	0.44
60.43	0.41
65.43	0.38
70.43	0.36
80.87	0.31
85.87	0.28
90.87	0.27
95.87	0.25
100.87	0.23
105.87	0.21
110.87	0.2
115.87	0.18
120.87	0.17
125.87	0.16
130.87	0.15
135.87	0.14
140.87	0.13
145.87	0.12

Parameter	Value
Construction Period	0.43
Service Life	75
Interval	5

Parameter	Value
Total	11.17

Parameter	Value
Total Routine Maintenance Costs	789,281.8
Total Routine Carbon Costs	126,924.07

Parameter	Value
Total Maintenance Costs	789,281.8
Total Carbon Costs	126,924.07

5.2.3 Major Inspection Costs

Breakdown

Major Inspection Cost

This calculates major inspection cost.

Equation

$$C_{mi} = \frac{C_0 \times p_c}{100}$$

Substitution

$$C_{mi} = \frac{12\,843\,979.44 \times p_c}{100}$$

Result

$$C_{mi} = 64\,219.9$$

Abbreviations

Symbol	Meaning
C_{mi}	Major Inspection Cost
C_0	Initial Cost Of Construction
p_c	Percentage Of Initial Construction Cost

Present Value Of Major Inspection Costs

This converts major inspection costs into present-value terms using the applicable discounting factor.

Equation

$$MIC = C_{mi} \times PWF$$

Substitution

$$MIC = C_{mi} \times PWF$$

Abbreviations

Symbol	Meaning
MIC	Present Value Of Major Inspection Costs
C_{mi}	Major Inspection Cost
PWF	Present Worth Factor

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Percentage Of Initial Construction Cost For Inspection	0.5
Interval For Repair And Rehabitation In Years For Inspection	5

Computed Values

Computed Value	Result
Major Inspection Cost	64,219.9
Sum Of Present Worth Factor For Inspection	11.17

Major Inspection Present Worth Factor Breakdown

Breakdown

Year To Pwf

Parameter	Value
5.43	0.92
10.43	0.86
15.43	0.8
20.43	0.74
25.43	0.69
30.43	0.64
35.43	0.59
40.43	0.55
45.43	0.51
50.43	0.48
55.43	0.44
60.43	0.41
65.43	0.38
70.43	0.36
80.87	0.31
85.87	0.28
90.87	0.27
95.87	0.25
100.87	0.23
105.87	0.21
110.87	0.2
115.87	0.18
120.87	0.17
125.87	0.16
130.87	0.15
135.87	0.14
140.87	0.13
145.87	0.12

Parameter	Value
Construction Period	0.43
Service Life	75
Interval	5

Parameter	Value
Total	11.17

Parameter	Value
Total Major Inspection Costs	717,528.91

5.2.4 Major Repair Carbon And Road User Costs

Breakdown

Major Repair Cost

This calculates major repair cost.

Equation

$$MR = \frac{C_0 \times p_c}{100}$$

Substitution

$$MR = \frac{12\,843\,979.44 \times p_c}{100}$$

Result

$$MR = 1\,284\,397.94$$

Abbreviations

Symbol	Meaning
MR	Major Repair Cost
C_0	Initial Cost Of Construction
p_c	Percentage Of Initial Construction Cost

Major Repair Carbon Cost

This calculates major repair carbon cost.

Equation

$$MRC = \frac{C_{CO_2,0} \times p_{CO_2}}{100}$$

Substitution

$$MRC = \frac{2\,065\,434.91 \times 0.55}{100}$$

Result

$$MRC = 11\,359.89$$

Abbreviations

Symbol	Meaning
MRC	Major Repair Carbon Cost
$C_{CO_2,0}$	Cost Of Initial Carbon Emissions In
p_{CO_2}	Percentage Of Initial Carbon Emission Cost

Present Value Of Major Repair Costs

This converts major repair costs into present-value terms using the applicable discounting factor.

Equation

$$MRC = MR \times PWF$$

Substitution

$$MRC = MR \times PWF$$

Abbreviations

Symbol	Meaning
MRC	Present Value Of Major Repair Costs
MR	Major Repair Cost
PWF	Present Worth Factor

Present Value Of Major Repair Carbon Costs

This converts major repair carbon costs into present-value terms using the applicable discounting factor.

Equation

$$MRCC = MRC \times PWF$$

Substitution

$$MRCC = MRC \times PWF$$

Abbreviations

Symbol	Meaning
$MRCC$	Present Value Of Major Repair Carbon Costs
MRC	Major Repair Carbon Cost
PWF	Present Worth Factor

Road User Cost

This calculates road user cost.

Formula note. calculated based on disruption duration and daily road user cost.

Result. Road User Cost = 26,323,141.21.

Vehicular Emission Cost

This calculates vehicular emission cost.

Formula note. calculated based on disruption duration and daily vehicular emission cost.

Result. Vehicular Emission Cost = 1,211,636.67.

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Cost Of Initial Carbon Emissions In	2,065,434.91
Percentage Of Initial Construction Cost For Repair	10
Percentage Of Initial Carbon Emission Cost	0.55
Interval For Repair And Rehabitation In Years For Repair	20
Repairs Duration Months	3
Disruption Days	90

Computed Values

Computed Value	Result
Major Repair Cost	1,284,397.94
Major Repair Carbon Cost	11,359.89
Sum Of Present Worth Factor For Repair	2.27
Road User Cost	26,323,141.21
Vehicular Emission Cost	1,211,636.67

Major Repair Present Worth Factor Breakdown

Breakdown

Year To Pwf

Parameter	Value
20.43	0.74
40.43	0.55
60.43	0.41
95.87	0.25
115.87	0.18
135.87	0.14

Parameter	Value
Construction Period	0.43
Service Life	75
Interval	20

Parameter	Value
Total	2.27

Road User Cost Breakdown

Parameter	Value
Daily Ruc	128,618.89
Duration Days	90
Spwf Applied	2.27

Sum Of Present Worth Factor Breakdown

Breakdown

Year To Pwf

Parameter	Value
20.43	0.74
40.43	0.55
60.43	0.41
95.87	0.25
115.87	0.18
135.87	0.14

Parameter	Value
Construction Period	0.43
Service Life	75
Interval	20

Parameter	Value
Total	2.27

Parameter	Value
Total Major Repair Costs	2,920,720.92
Total Major Repair Carbon Costs	25,832.39
Total Major Repair Road User Costs	26,323,141.21
Total Major Repair Vehicular Emission Costs	1,211,636.67

5.2.5 Replacement Costs For Bearing And Expansion Joint

Breakdown

Replacement Cost

This calculates replacement cost.

Equation

$$C_{rep} = \frac{C_{ss} \times p_{ss}}{100}$$

Substitution

$$C_{rep} = \frac{9\,356\,038.92 \times 12.5}{100}$$

Result

$$C_{rep} = 1\,169\,504.86$$

Abbreviations

Symbol	Meaning
C_{rep}	Replacement Cost
C_{ss}	Cost Of Super Structure
p_{ss}	Percentage Of Super Structure Cost

Present Value Of Replacement Costs

This converts replacement costs into present-value terms using the applicable discounting factor.

Equation

$$RC = C_{rep} \times PWF$$

Substitution

$$RC = C_{rep} \times PWF$$

Abbreviations

Symbol	Meaning
RC	Present Value Of Replacement Costs
C_{rep}	Replacement Cost
PWF	Present Worth Factor

Road User Cost

This calculates road user cost.

Formula note. calculated based on duration of replacement and daily road user cost.

Result. Road User Cost = 399,747.5.

Vehicular Emission Cost

This calculates vehicular emission cost.

Formula note. calculated based on duration of replacement and daily vehicular emission cost.

Result. Vehicular Emission Cost = 18,400.11.

Inputs

Input	Value
Cost Of Super Structure	9,356,038.92
Percentage Of Super Structure Cost	12.5
Interval For Replacement In Years	25
Duration Of Replacement In Days	2

Computed Values

Computed Value	Result
Replacement Cost	1,169,504.86
Sum Of Present Worth Factor For Replacement	1.55
Road User Cost	399,747.5
Vehicular Emission Cost	18,400.11

Present Worth Factor Breakdown

Breakdown

Year To Pwf

Parameter	Value
25.43	0.69
50.43	0.48
100.87	0.23
125.87	0.16

Parameter	Value
Construction Period	0.43
Service Life	75
Interval	25

Parameter	Value
Total	1.55

Road User Cost Breakdown

Parameter	Value
Daily Ruc	128,618.89
Duration Days	2
Spwf Applied	1.55

Parameter	Value
Total Replacement Costs	1,817,410.56
Total Replacement Road User Costs	399,747.5
Total Replacement Vehicular Emission Costs	18,400.11

5.3 Reconstruction Detailed Calculations

5.3.1 Present Worth Factor For Demolition

Parameter	Value
75.52	0.33

5.3.2 Demolition And Disposal Breakdown

Demolition Cost

This calculates demolition cost.

Equation

$$C_{demo} = \frac{C_0 \times p_c}{100}$$

Substitution

$$C_{demo} = \frac{12\,843\,979.44 \times 10}{100}$$

Result

$$C_{demo} = 1\,284\,397.94$$

Abbreviations

Symbol	Meaning
C_{demo}	Demolition Cost
C_0	Initial Cost Of Construction
p_c	Percentage Of Initial Construction Cost

Demolition Carbon Cost

This calculates demolition carbon cost.

Equation

$$C_{CO_2,demo} = \frac{C_{CO_2,0} \times p_{CO_2}}{100}$$

Substitution

$$C_{CO_2,demo} = \frac{2\,065\,434.91 \times 10}{100}$$

Result

$$C_{CO_2,demo} = 206\,543.49$$

Abbreviations

Symbol	Meaning
$C_{CO_2,demo}$	Demolition Carbon Cost
$C_{CO_2,0}$	Cost Of Initial Carbon Emissions In
p_{CO_2}	Percentage Of Initial Carbon Emission Cost

Cost Of Reconstruction After Demolition

This calculates cost of reconstruction after demolition.

Equation

$$C_{recon} = C_0$$

Substitution

$$C_{recon} = 12\,843\,979.44$$

Abbreviations

Symbol	Meaning
C_{recon}	Cost Of Reconstruction After Demolition
C_0	Initial Cost Of Construction

Present Value Of Demolition Costs

This converts demolition costs into present-value terms using the applicable discounting factor.

Equation

$$DC = C_{demo} \times PWF_{demo}$$

Substitution

$$DC = C_{demo} \times PWF_{demo}$$

Abbreviations

Symbol	Meaning
DC	Present Value Of Demolition Costs
C_{demo}	Demolition Cost
PWF_{demo}	Demolition Spwi

Present Value Of Demolition Carbon Costs

This converts demolition carbon costs into present-value terms using the applicable discounting factor.

Equation

$$DCC = C_{CO_2,demo} \times PWF_{demo}$$

Substitution

$$DCC = C_{CO_2,demo} \times PWF_{demo}$$

Abbreviations

Symbol	Meaning
DCC	Present Value Of Demolition Carbon Costs
$C_{CO_2,demo}$	Demolition Carbon Cost
PWF_{demo}	Demolition Spwi

Present Value Of Reconstruction Costs

This converts reconstruction costs into present-value terms using the applicable discounting factor.

Equation

$$RC = C_0 \times PWF_{demo}$$

Substitution

$$RC = 12\,843\,979.44 \times PWF_{demo}$$

Abbreviations

Symbol	Meaning
RC	Present Value Of Reconstruction Costs
C_0	Initial Cost Of Construction
PWF_{demo}	Demolition Spwi

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Cost Of Initial Carbon Emissions In	2,065,434.91
Percentage Of Initial Construction Cost	10
Percentage Of Initial Carbon Emission Cost	10

Computed Values

Computed Value	Result
Demolition Cost	1,284,397.94
Demolition Carbon Cost	206,543.49
Demolition Spwi	0.33

5.3.3 Road User And Vehicular Emission Costs Breakdown

Demolition Road User Cost Breakdown

Parameter	Value
Daily Ruc	128,618.89
Duration Days	30
Spwf Applied	0.33

Demolition Vehicular Emission Cost Breakdown

Parameter	Value
Daily Ruc	128,618.89
Duration Days	30
Spwf Applied	0.33

Parameter	Value
Demolition Road User Cost	1,277,185.54
Demolition Vehicular Emission Cost	58,788

5.3.4 Reconstruction Costs Breakdown

Present Value Of Construction Costs

This converts construction costs into present-value terms using the applicable discounting factor.

Equation

$$CC = C_0 \times SPWF$$

Substitution

$$CC = 12\,843\,979.44 \times 0.33$$

Result

$$CC = 4\,251\,357.19$$

Abbreviations

Symbol	Meaning
CC	Present Value Of Construction Costs
C_0	Initial Cost Of Construction
$SPWF$	Sum Of Present Worth Factor

Present Value Of Carbon Costs

This converts carbon costs into present-value terms using the applicable discounting factor.

Equation

$$CC = C_{CO_2,0} \times SPWF$$

Substitution

$$CC = 2\,065\,434.91 \times 0.33$$

Result

$$CC = 683\,658.96$$

Abbreviations

Symbol	Meaning
CC	Present Value Of Carbon Costs
$C_{CO_2,0}$	Cost Of Initial Carbon Emissions In
$SPWF$	Sum Of Present Worth Factor

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Cost Of Initial Carbon Emissions In	2,065,434.91
Sum Of Present Worth Factor	0.33

Computed Values

Computed Value	Result
Present Value Of Construction Costs	4,251,357.19
Present Value Of Carbon Costs	683,658.96
Construction Road User Cost	6,641,364.8
Construction Vehicular Emission Cost	305,697.6
Construction Road User Cost Breakdown	{'daily_ruc': 128618.89, 'duration_days': 156, 'spwf_applied': 0.331}
Construction Vehicular Emission Cost Breakdown	{'daily_ruc': 128618.89, 'duration_days': 156, 'spwf_applied': 0.331}

5.3.5 Total Demolition And Disposal Costs

425,135.72

5.3.6 Total Demolition And Disposal Carbon Costs

68,365.9

5.3.7 Cost Of Reconstruction After Demolition

4,251,357.19

5.3.8 Carbon Cost Of Reconstruction After Demolition

683,658.96

5.3.9 Ruc Demolition

1,277,185.54

5.3.10 Ruc Reconstruction

6,641,364.8

5.3.11 Demolition Vehicular Emission Cost

58,788

5.3.12 Reconstruction Vehicular Emission Cost

305,697.6

5.3.13 Time Cost Of Loan

Time Cost Of Loan

This calculates time cost of loan.

Equation

$$C_{loan} = C_0 \times r \times t_c \times I \times SPWF$$

Substitution

$$C_{loan} = 12\,843\,979.44 \times 0.08 \times 0.43 \times 0.5 \times 0.33$$

Result

$$C_{loan} = 71\,387.37$$

Abbreviations

Symbol	Meaning
C_{loan}	Time Cost Of Loan
C_0	Initial Cost Of Construction
r	Interest Rate
t_c	Time For Construction Years
I	Investment Ratio
$SPWF$	Sum Of Present Worth Factor

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Interest Rate	0.08
Time For Construction Years	0.43
Investment Ratio	0.5
Sum Of Present Worth Factor	0.33

Computed Values

Computed Value	Result
Time Cost Of Loan	71,387.37

5.3.14 Total Scrap Value

Total Scrap Value

This calculates total scrap value.

Equation

$$SCRA = S \times PWF_{demo}$$

Result

$$SCRA = 716\,315.45$$

Abbreviations

Symbol	Meaning
$SCRA$	Total Scrap Value
S	Total Scrap Cost
PWF_{demo}	Demolition Spwi

Note. This value is negative as it represents a salvage value..

Parameter	Value
Value	716,315.45

5.4 End-of-Life Stage Detailed Calculations

5.4.1 Present Worth Factor For Demolition

Parameter	Value
150	0.11

5.4.2 Demolition And Disposal Breakdown

Demolition Cost

This calculates demolition cost.

Equation

$$C_{demo} = \frac{C_0 \times p_c}{100}$$

Substitution

$$C_{demo} = \frac{12\,843\,979.44 \times 10}{100}$$

Result

$$C_{demo} = 1\,284\,397.94$$

Abbreviations

Symbol	Meaning
C_{demo}	Demolition Cost
C_0	Initial Cost Of Construction
p_c	Percentage Of Initial Construction Cost

Demolition Carbon Cost

This calculates demolition carbon cost.

Equation

$$C_{CO_2,demo} = \frac{C_{CO_2,0} \times p_{CO_2}}{100}$$

Substitution

$$C_{CO_2,demo} = \frac{2\,065\,434.91 \times 10}{100}$$

Result

$$C_{CO_2,demo} = 206\,543.49$$

Abbreviations

Symbol	Meaning
$C_{CO_2,demo}$	Demolition Carbon Cost
$C_{CO_2,0}$	Cost Of Initial Carbon Emissions In
p_{CO_2}	Percentage Of Initial Carbon Emission Cost

Cost Of Reconstruction After Demolition

This calculates cost of reconstruction after demolition.

Equation

$$C_{recon} = C_0$$

Substitution

$$C_{recon} = 12\,843\,979.44$$

Abbreviations

Symbol	Meaning
C_{recon}	Cost Of Reconstruction After Demolition
C_0	Initial Cost Of Construction

Present Value Of Demolition Costs

This converts demolition costs into present-value terms using the applicable discounting factor.

Equation

$$DC = C_{demo} \times PWF_{demo}$$

Substitution

$$DC = C_{demo} \times PWF_{demo}$$

Abbreviations

Symbol	Meaning
DC	Present Value Of Demolition Costs
C_{demo}	Demolition Cost
PWF_{demo}	Demolition Spwi

Present Value Of Demolition Carbon Costs

This converts demolition carbon costs into present-value terms using the applicable discounting factor.

Equation

$$DCC = C_{CO_2,demo} \times PWF_{demo}$$

Substitution

$$DCC = C_{CO_2,demo} \times PWF_{demo}$$

Abbreviations

Symbol	Meaning
DCC	Present Value Of Demolition Carbon Costs
$C_{CO_2,demo}$	Demolition Carbon Cost
PWF_{demo}	Demolition Spwi

Present Value Of Reconstruction Costs

This converts reconstruction costs into present-value terms using the applicable discounting factor.

Equation

$$RC = C_0 \times PWF_{demo}$$

Substitution

$$RC = 12\,843\,979.44 \times PWF_{demo}$$

Abbreviations

Symbol	Meaning
RC	Present Value Of Reconstruction Costs
C_0	Initial Cost Of Construction
PWF_{demo}	Demolition Spwi

Inputs

Input	Value
Initial Cost Of Construction	12,843,979.44
Cost Of Initial Carbon Emissions In	2,065,434.91
Percentage Of Initial Construction Cost	10
Percentage Of Initial Carbon Emission Cost	10

Computed Values

Computed Value	Result
Demolition Cost	1,284,397.94
Demolition Carbon Cost	206,543.49
Demolition Spwi	0.11

5.4.3 Total Demolition And Disposal Costs

142,568.17

5.4.4 Total Demolition And Disposal Carbon Costs

22,926.33

5.4.5 Cost Of Reconstruction After Demolition

1,425,681.72

5.4.6 Ruc Demolition

428,300.89

5.4.7 Demolition Vehicular Emission Cost

19,714.4

5.4.8 Total Scrap Value

Total Scrap Value

This calculates total scrap value.

Equation

$$SCRA = S \times PWF_{demo}$$

Result

$$SCRA = 240\,214.55$$

Abbreviations

Symbol	Meaning
$SCRA$	Total Scrap Value
S	Total Scrap Cost
PWF_{demo}	Demolition Spwi

Parameter	Value
Value	240,214.55

6 Summary

Stage	Total
Initial Stage	36,113,190.09
Use Stage	35,113,145.53
Reconstruction	14,499,256.53
End-of-Life Stage	853,724.34
Grand Total	86,579,316.49